

Multi-Sensor Fusion for Autonomous Driving Perception: A Survey

Multi-sensor fusion is a core problem in autonomous driving perception because cameras, LiDAR, radar, maps, and temporal context provide complementary evidence about dynamic road scenes. Cameras provide dense texture and semantics, LiDAR provides accurate geometry, radar provides range and velocity robustness, and temporal fusion stabilizes perception over time. This survey synthesizes an arXiv-derived corpus of 300 papers collected during this run and organizes the literature around sensor architectures, BEV fusion, 3D detection, occupancy and mapping, temporal fusion, calibration, robustness, datasets, and safety. The survey argues that modern fusion is less about concatenating features and more about building a consistent spatiotemporal representation under sensor noise, missing modalities, calibration errors, and adverse weather. We close with prospects for uncertainty-aware fusion, foundation models for driving perception, and evaluation under real deployment shifts.

Keywords:

multi-sensor fusion; autonomous driving; perception; camera-LiDAR fusion; radar; BEV perception; 3D object detection; occupancy

1 Introduction

Autonomous driving perception depends on sensors with complementary strengths. Cameras capture dense appearance, semantic cues, traffic lights, lane markings, and long-range context. LiDAR provides accurate metric geometry and object shape. Radar offers velocity and robustness under rain, fog, and darkness. Maps, inertial sensors, and temporal history provide additional structure. Multi-sensor fusion studies how these signals should be calibrated, synchronized, aligned, and combined for reliable perception [23, 30, 43, 117, 153, 189, 210, 265].

The field has moved from early object-level fusion to feature-level and BEV-centric fusion. Bird’s-eye-view representations are attractive because they align camera, LiDAR, radar, map, and planning coordinates in a common road-centric frame [23, 30, 43, 117, 153, 210, 254, 265]. At the same time, fusion remains difficult: sensors have different resolution, field of view, latency, failure modes, and uncertainty. A good fusion system should improve perception when all sensors work, degrade gracefully when

one modality is missing, and remain robust under domain shifts.

This survey reviews multi-sensor fusion for autonomous driving perception. We focus on camera-LiDAR-radar architectures, BEV fusion, 3D detection, occupancy, temporal fusion, calibration, robustness, datasets, and evaluation.

2 Background

2.1 Sensor Modalities

Cameras, LiDAR, and radar encode different aspects of the driving scene. Cameras provide semantics but lack direct depth; LiDAR provides sparse but accurate range; radar provides noisy but robust velocity and range. Sensor fusion aims to combine these properties without importing each modality’s weaknesses into the final representation [23, 30, 43, 114, 153, 176, 210, 237].

2.2 Coordinate Frames and Alignment

Fusion requires geometric alignment among image planes, point clouds, radar returns, vehicle coordinates, and map coordinates. Errors in calibration or synchronization can cause inconsistent features and false detections. Modern systems therefore often learn alignment implicitly while still depending on accurate sensor metadata [11, 48, 65, 76, 171, 207, 210, 297].

2.3 Perception Tasks

Autonomous driving perception includes 3D detection, tracking, semantic and panoptic segmentation, lane detection, occupancy prediction, free-space estimation, motion forecasting, and scene understanding. Fusion strategies differ by task: detection needs precise object localization, occupancy needs dense spatial reasoning, and tracking needs temporal consistency [23, 30, 43, 117, 153, 189, 210, 265].

3 Methods

3.1 Search Protocol and Corpus Construction

The corpus for this survey was generated during this run from the arXiv structured API. Queries combined the task topic with multi-sensor fusion, camera-LiDAR fusion, radar-camera fusion, autonomous driving perception, 3D object detection, BEV fusion, BEVFusion, BEVFormer, TransFusion, CenterFusion, occupancy prediction, semantic scene completion, nuScenes, Waymo, KITTI, Argoverse, temporal fusion, calibration, missing modality, adverse weather, robust sensor fusion, and cooperative perception. Records were retained when title or abstract evidence connected autonomous driving or self-driving perception with sensor fusion, camera/LiDAR/radar integration, BEV representations, detection, occupancy, tracking, calibration, robustness, datasets, or evaluation. Duplicates were removed by arXiv identifier, and the final bibliography cites 300 unique scholarly preprints.

3.2 Taxonomy

We organize the field along six axes. The first is *sensor set*: camera-only with depth cues, camera-LiDAR, camera-radar, LiDAR-radar, camera-LiDAR-radar, map-enhanced, or cooperative multi-agent sensing. The second is *fusion level*: raw data, proposal, feature, BEV, object, track, or

decision fusion. The third is *representation*: image plane, point cloud, voxel grid, pillar grid, BEV feature map, occupancy grid, or temporal memory. The fourth is *task*: detection, segmentation, occupancy, tracking, forecasting, mapping, or planning-oriented perception. The fifth is *robustness*: calibration error, sensor dropout, weather, domain shift, adversarial noise, and uncertainty. The sixth is *evaluation*: accuracy, latency, calibration sensitivity, missing-modality behavior, and safety relevance.

3.3 Camera-LiDAR-Radar Fusion Architectures

Fusion architectures differ in where modalities meet. Early fusion projects point clouds into image space or images into 3D grids; middle fusion aligns learned features; late fusion combines detections or tracks. Radar adds sparse but useful velocity evidence, while LiDAR often anchors geometry. Modern designs increasingly use attention, transformers, and BEV feature aggregation to fuse modalities in a common representation [23, 30, 43, 114, 153, 176, 210, 237].

3.4 BEV Fusion and Transformers

BEV fusion converts perspective images, point clouds, and radar returns into a road-plane representation. This makes multi-camera fusion, map interaction, temporal memory, and planning easier. Transformer-based systems use cross-attention to query image features from BEV cells or fuse LiDAR and camera features. The challenge is preserving height, occlusion, and uncertainty while keeping computation tractable [23, 30, 43, 117, 153, 210, 254, 265].

3.5 3D Detection and Semantic Perception

3D object detection remains the most studied fusion task. Fusion can improve localization, classification, velocity estimation, and long-range detection. Semantic segmentation and lane perception benefit from image detail and LiDAR geometry. A persistent issue is that benchmark improvements may hide failures under rare weather, unusual objects, or miscalibrated sensors [23, 30, 43, 117, 153, 189, 210, 265].

3.6 Occupancy, Mapping, and Scene Understanding

Occupancy prediction and semantic scene completion move beyond boxes toward dense understanding

of free, occupied, and unknown space. Fusion is essential because cameras infer semantics while LiDAR and radar provide geometric evidence. Occupancy representations also connect perception to planning by providing a spatial substrate for future motion reasoning [54, 109, 125, 152, 160, 210, 265, 292].

3.7 Temporal Fusion, Tracking, and Prediction

Driving perception is sequential. Temporal fusion aggregates multi-frame evidence, improves detection under occlusion, stabilizes velocity estimates, and supports tracking and forecasting. However, temporal fusion also propagates errors when ego-motion, calibration, or association is wrong. Good systems need uncertainty-aware memory and mechanisms for forgetting stale evidence [11, 23, 146, 152, 210, 213, 265, 287].

3.8 Calibration, Synchronization, and Uncertainty

Sensor fusion depends on extrinsic calibration, timing, and uncertainty estimation. Small errors can misalign image features and point clouds, especially for distant objects or fast motion. Research on self-calibration, missing modality, confidence estimation, and uncertainty-aware fusion addresses these deployment concerns [11, 48, 65, 76, 171, 207, 210, 297].

3.9 Robustness and Safety

Autonomous vehicles must operate under rain, fog, snow, night, glare, sensor blockage, and domain shifts. Fusion can increase robustness by letting modalities compensate for one another, but it can also fail if the system over-trusts a corrupted modality. Robust fusion should explicitly model sensor reliability and report performance under stress tests [11, 43, 114, 117, 146, 176, 210, 265].

3.10 Datasets, Benchmarks, and Evaluation

Datasets such as KITTI, nuScenes, Waymo, Argoverse, and related benchmarks shaped the field by standardizing sensor configurations and metrics. Yet benchmark scores do not fully capture calibration drift, missing sensors, weather, latency, or closed-loop planning impact. Evaluation should therefore include stress conditions and deployment-relevant metrics [23, 30, 43, 117, 153, 210, 237, 265].

3.11 Representative Literature Map

The following map cites the full retained corpus by theme, making the survey coverage explicit.

Camera-lidar-radar fusion architectures The camera-lidar-radar fusion architectures cluster contains work that develops methods, systems, datasets, or analyses central to this survey [2, 3, 94, 117, 162, 180, 189, 242, 292]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [26, 34, 75, 132, 201, 205, 218, 284, 285]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [84, 104, 123, 134, 155, 178, 229, 271, 296]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [1, 8, 64, 74, 126, 199].

Bev representations and transformer fusion The bev representations and transformer fusion cluster contains work that develops methods, systems, datasets, or analyses central to this survey [23, 30, 107, 138, 153, 187, 213, 237, 254]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [32, 86, 90, 133, 139, 156, 175, 186, 297]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [52, 83, 130, 168, 190, 208, 280, 282, 290]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [22, 36, 39, 144, 157, 188, 224, 226, 289]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [131, 251, 258].

3d object detection and semantic perception The 3d object detection and semantic perception cluster contains work that develops methods, systems, datasets, or analyses central to this survey [57, 65, 111, 112, 164, 176, 259, 267, 288]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [38, 50, 142, 194, 196, 217, 245, 252, 253]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [12, 77, 119, 141, 143, 161, 166, 182, 249]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [17, 25, 89, 92, 96, 184, 203, 234, 250]. Additional

retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [6, 58, 202, 206, 236, 246, 247, 264, 273]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [262, 274].

Occupancy, mapping, and scene understanding The occupancy, mapping, and scene understanding cluster contains work that develops methods, systems, datasets, or analyses central to this survey [160].

Temporal fusion, tracking, and prediction The temporal fusion, tracking, and prediction cluster contains work that develops methods, systems, datasets, or analyses central to this survey [54, 63, 68, 69, 152, 191, 241, 276, 287]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [13, 67, 124, 173, 183, 195, 278, 300].

Calibration, synchronization, and uncertainty The calibration, synchronization, and uncertainty cluster contains work that develops methods, systems, datasets, or analyses central to this survey [181].

Robustness, adverse weather, and safety The robustness, adverse weather, and safety cluster contains work that develops methods, systems, datasets, or analyses central to this survey [43, 60, 73, 78, 109, 114, 154, 231, 232]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific variants [40, 48, 59, 135, 212, 223, 260, 277, 299]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [4, 20, 41, 97, 140, 219, 238, 263, 294]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [27, 49, 71, 79, 100, 127, 145, 197, 257]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [5, 14, 88, 137, 227, 281].

Datasets, benchmarks, and evaluation The datasets, benchmarks, and evaluation cluster contains work that develops methods, systems, datasets, or analyses central to this survey [11, 19, 76, 80, 98, 146, 148, 210, 265]. A second strand in this cluster studies design trade-offs, evaluation protocols, failure modes, and domain-specific

variants [28, 103, 106, 125, 171, 185, 225, 255, 275]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [10, 66, 149, 193, 207, 239, 248, 268, 295]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [45, 82, 113, 174, 177, 211, 220, 240, 243]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [9, 16, 121, 128, 163, 167, 216, 230, 293]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [44, 62, 99, 115, 150, 170, 172, 192, 235]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [15, 21, 24, 29, 122, 136, 221, 228, 269]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [46, 51, 56, 87, 120, 198, 215, 270, 272]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [53, 61, 93, 95, 101, 105, 110, 151, 266]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [18, 35, 55, 72, 81, 147, 158, 159, 179]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [7, 31, 33, 37, 70, 91, 118, 165, 233]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [47, 169, 200, 209, 214, 244, 283, 291, 298]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [85, 102, 108, 116, 129, 204, 222, 261, 279]. Additional retained studies broaden the cluster with alternative architectures, training signals, data sources, and deployment assumptions [42, 256, 286].

4 Challenges and Prospects

4.1 Alignment Under Real-World Noise

Fusion assumes aligned sensor streams, but real vehicles experience calibration drift, latency, rolling shutter, and motion distortion. Future systems should estimate alignment uncertainty and expose when fused evidence is unreliable.

4.2 Graceful Degradation

A robust autonomous vehicle should continue operating when a sensor is degraded or missing. This requires training and evaluation protocols for missing modalities, corrupted inputs, and weather shifts rather than only ideal benchmark data.

4.3 From Detection to Occupancy and Planning

The field is moving from object boxes toward dense occupancy, map-aware perception, and planning-oriented representations. Fusion methods should be evaluated by how well they support downstream decision making, not only detection metrics.

4.4 Foundation Models and Open-World Perception

Large vision and multimodal models may improve semantic coverage, but driving perception requires metric geometry, latency, calibration, and safety. Future systems will likely combine foundation-model semantics with sensor-grounded BEV and occupancy representations.

5 Conclusion

Multi-sensor fusion is central to autonomous driving perception because no single sensor is sufficient across all road conditions. This survey reviewed camera-LiDAR-radar fusion, BEV transformers, 3D detection, occupancy, temporal fusion, calibration, robustness, benchmarks, and safety. The central lesson is that fusion must be treated as calibrated, uncertainty-aware spatiotemporal reasoning. Progress will depend on robust alignment, graceful degradation, dense scene representations, and evaluation under real deployment shifts.

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